

MATHEMATICS**Maximum Marks: 80****Time Allowed: Three Hours***Reading Time: Additional Fifteen Minutes***Instructions to Candidates**

1. You are allowed an additional **fifteen minutes** for **only** reading the paper.
2. You must **NOT** start writing during reading time.
3. The question paper has **9** printed pages.
4. It consists of **twenty** questions and **four** sections: Sections A, B, C and D.
5. All questions are compulsory.
6. Section A consists of very short answer questions of 1 mark each.
7. Section B consists of short answer questions of 2 marks each.
8. Section C consists of moderately long answer questions of 3 marks each.
9. Section D consists of long answer questions of 5 marks each.
10. Internal choices have been provided in **three** questions, each in Sections B, C and D.
11. While attempting Multiple Choice Questions in Section A, you are required to write only **ONE** option as the answer.
12. The intended marks for questions or parts of questions are given in the brackets [].
13. All workings, including rough work, should be done on the same page as, and adjacent to, the rest of the answer.
14. Mathematical tables and graph papers are provided.

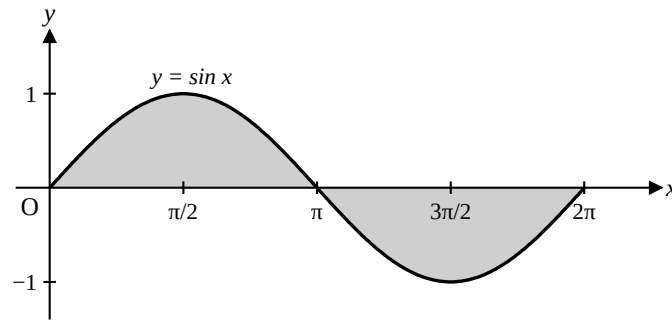
Instruction to Supervising Examiner*Kindly read aloud the instructions given above to all the candidates present in the examination hall.***SECTION A – 20 MARKS****Question 1**

In subparts (i) to (xvii) choose the correct options and in subparts (xviii) to (xx), answer the questions as instructed.

- (i) The relation $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1), (2, 3), (3, 2)\}$ on the set $A = \{1, 2, 3\}$ is: [1]
(Understand)
- (a) an equivalence relation
 - (b) reflexive and symmetric but not transitive
 - (c) reflexive and transitive but not symmetric
 - (d) symmetric and transitive but not reflexive

- (ii) For $x \in \mathbb{R}$, the expression $\tan^{-1}x + 2 \cot^{-1}x$ lies in the interval: *(Analyse)* [1]
- (a) $(0, \pi)$
 (b) $(\frac{\pi}{2}, \frac{3\pi}{2})$
 (c) $(-\frac{\pi}{2}, \frac{\pi}{2})$
 (d) $[\frac{\pi}{2}, \frac{3\pi}{2}]$
- (iii) The area of the triangle with vertices $(k, 0)$, $(4, 0)$ and $(0, 2)$ is 4 square units. Then k is: *(Apply)* [1]
- (a) 0 only
 (b) 8 only
 (c) 0 or 8
 (d) -8 or 8
- (iv) The volume of a spherical balloon is increasing at $25 \text{ cm}^3/\text{s}$. The rate of increase of its surface area when the radius is 5 cm is: *(Apply)* [1]
- (a) $5 \text{ cm}^2/\text{s}$
 (b) $10 \text{ cm}^2/\text{s}$
 (c) $20 \text{ cm}^2/\text{s}$
 (d) $25 \text{ cm}^2/\text{s}$
- (v) **Statement I:** The function $f(x) = e^x + e^{-x}$ is strictly decreasing on $(-\infty, 0)$. [1]
Statement II: The minimum value of $f(x) = e^x + e^{-x}$ on $\mathbb{R}$ is 2.
(Understand)
- (a) Statement I is true and Statement II is false.
 (b) Statement I is false and Statement II is true.
 (c) Both the statements are true.
 (d) Both the statements are false.
- (vi) The equation of the tangent to the curve $y = x^2 - 2x + 7$ which is parallel to the line $2x - y + 9 = 0$ is: *(Apply)* [1]
- (a) $y = 2x + 3$
 (b) $y = 2x - 3$
 (c) $y = 2x + 7$
 (d) $y = -2x + 11$
- (vii) The absolute maximum and absolute minimum values of $f(x) = x^3 - 3x$ on $[0, 2]$ are: *(Apply)* [1]
- (a) 2 and -2
 (b) 0 and -2
 (c) 2 and 0
 (d) 18 and -2

- (viii) The total area of the shaded region between the curve $y = \sin x$ and the x-axis, $0 \leq x \leq 2\pi$, is given by: (*Apply*) [1]



- (a) $\int_0^{2\pi} \sin x \, dx$
 (b) $2 \int_0^{\pi} \sin x \, dx$
 (c) $\int_0^{\pi} \sin x \, dx + \int_{\pi}^{2\pi} \sin x \, dx$
 (d) $\int_0^{\pi} \sin x \, dx$
- (ix) A student evaluates $\int x \cos x \, dx$ as follows. [1]
 Step 1: Take $u = x$, $dv = \cos x \, dx$, so that $du = dx$ and $v = -\sin x$.
 Step 2: $\int x \cos x \, dx = -x \sin x + \int \sin x \, dx$.
 Step 3: $= -x \sin x - \cos x + C$.
 In which step is there an error (if any)? (*Understand*)
 (a) Step 1
 (b) Step 2
 (c) Step 3
 (d) There is no error
- (x) Two students evaluate $\int \frac{dx}{x^2 + 2x + 2}$. Asha writes $\tan^{-1}(x + 1) + C$ and Ben writes $\log |x^2 + 2x + 2| + C$. Whose answer is correct? (*Understand*) [1]
 (a) Only Asha's
 (b) Only Ben's
 (c) Both are correct
 (d) Neither is correct
- (xi) The area of the parallelogram whose adjacent sides are $\hat{i} + 2\hat{j} + 3\hat{k}$ and $3\hat{i} - 2\hat{j} + \hat{k}$ is: (*Apply*) [1]
 (a) $8\sqrt{3}$ sq units
 (b) $4\sqrt{3}$ sq units
 (c) $16\sqrt{3}$ sq units
 (d) 64 sq units

- (xii) The distance of the point $(2, 3, -5)$ from the plane $x + 2y - 2z = 9$ is: (*Apply*) [1]
- (a) 3 units
 (b) 9 units
 (c) 1 unit
 (d) $\frac{13}{3}$ units
- (xiii) The corner points of the feasible region of an LPP are $(0, 0)$, $(4, 0)$, $(3, 2)$ and $(0, 4)$. If the maximum of $Z = px + qy$ ($p, q > 0$) occurs at both $(4, 0)$ and $(3, 2)$, then: (*Apply*) [1]
- (a) $p = q$
 (b) $p = 2q$
 (c) $q = 2p$
 (d) $p = 3q$
- (xiv) **Statement I:** The origin lies in the half-plane $2x + 3y \leq 6$. [1]
Statement II: The system $x + y \geq 4$, $x + y \leq 2$, $x \geq 0$, $y \geq 0$ has no feasible region, so an LPP with these constraints has no optimal solution.
 (*Understand*)
- (a) Statement I is true and Statement II is false.
 (b) Statement I is false and Statement II is true.
 (c) Both the statements are true.
 (d) Both the statements are false.
- (xv) Two fair dice are thrown. Given that the first die shows an odd number, the probability that the sum of the numbers is 8 is: (*Apply*) [1]
- (a) $\frac{1}{9}$
 (b) $\frac{1}{18}$
 (c) $\frac{5}{36}$
 (d) $\frac{1}{6}$
- (xvi) **Assertion:** The number of one-one functions from $A = \{1, 2, 3\}$ to $B = \{a, b\}$ is 0. [1]
Reason: If $n(A) > n(B)$, there is no one-one function from A to B.
 (*Understand*)
- (a) Both Assertion and Reason are true and Reason is the correct explanation for Assertion.
 (b) Both Assertion and Reason are true but Reason is not the correct explanation for Assertion.
 (c) Assertion is true and Reason is false.
 (d) Assertion is false and Reason is true.

(xvii) **Assertion:** $\int_0^{\frac{\pi}{2}} \log (\tan x) dx = 0.$ [1]

Reason: $\int_0^a f(x) dx = \int_0^a f(a-x) dx.$

(Understand)

- (a) Both Assertion and Reason are true and Reason is the correct explanation for Assertion.
- (b) Both Assertion and Reason are true but Reason is not the correct explanation for Assertion.
- (c) Assertion is true and Reason is false.
- (d) Assertion is false and Reason is true.

(xviii) If A is a square matrix such that $A^2 = A$, find $(I + A)^2 - 3A.$ (Apply) [1]

(xix) Find the integrating factor of the differential equation $x \frac{dy}{dx} - y = x^2 \log x.$ (Understand) [1]

(xx) A random variable X has $P(X = x) = k(x + 1)$ for $x = 0, 1, 2, 3$ and 0 otherwise. Find $P(X \geq 2).$ (Apply) [1]

SECTION B – 14 MARKS

Question 2

Find the angle between the planes $2x - y + z = 6$ and $x + y + 2z = 7.$ (Apply) [2]

Question 3

A committee of 4 is to be chosen at random from 6 boys and 4 girls.

- (i) Find the probability that the committee has exactly 2 girls. (Apply) [1]
- (ii) Find the probability that it has at least 2 girls. (Evaluate) [1]

Question 4

Solve the differential equation $\frac{dy}{dx} + y \cot x = 2x + x^2 \cot x$ ($x \neq 0$), given that $y = 0$ when $x = \frac{\pi}{2}.$ (Apply) [2]

Question 5

Show that the function $f(x) = |x - 3|$ is continuous but not differentiable at $x = 3.$ (Understand) [2]

OR

If $x = a(\theta + \sin \theta)$ and $y = a(1 - \cos \theta)$, find $\frac{dy}{dx}$ at $\theta = \frac{\pi}{2}.$ (Apply) [2]

Question 6

Solve for x: $2 \tan^{-1}(\cos x) = \tan^{-1}(2 \operatorname{cosec} x), 0 < x < \pi.$ (Apply) [2]

OR

Evaluate: $\tan(2 \tan^{-1} \frac{1}{5} - \frac{\pi}{4}).$ (Apply) [2]

Question 7

If $|\vec{a}| = 3$, $|\vec{b}| = 4$ and $|\vec{a} + \vec{b}| = 5$, find $|\vec{a} - \vec{b}|$. (*Analyse*)

[2]

OR

Find a vector of magnitude 6 which is perpendicular to both $\vec{a} = 3\hat{i} + 2\hat{j} + 2\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} - 2\hat{k}$. (*Apply*)

[2]

Question 8

Three garden sprinklers are fixed at A(1, -1, 2), B(2, 1, -1) and C(3, -1, 2), with coordinates in metres.

(i) Find the cosine of the angle BAC. (*Apply*)

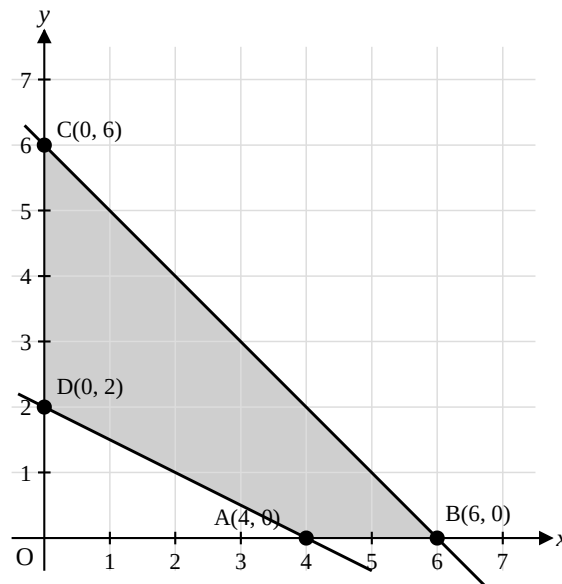
[1]

(ii) Find the area of the triangular patch ABC watered between them. (*Apply*)

[1]

SECTION C – 21 MARKS**Question 9**

The shaded region in the figure is the feasible region of a linear programming problem.



(i) Write the constraints that define the feasible region. (*Understand*)

[1]

(ii) Find the minimum value of $Z = 3x + 5y$ over the region. (*Apply*)

[1]

(iii) If $Z = px + qy$ ($p, q > 0$) takes its minimum value at both A and D, find the relation between p and q . (*Analyse*)

[1]

Question 10

If $y = (\tan^{-1}x)^2$, show that $(1 + x^2)^2 \frac{d^2y}{dx^2} + 2x(1 + x^2) \frac{dy}{dx} = 2$. (*Apply*)

[3]

Question 11

Let $f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$, $|x| < 1$.

(i) Using the substitution $x = \tan \theta$, show that $f'(x) = \frac{2}{1+x^2}$. (*Apply*)

[2]

- (ii) A student claims that f has a stationary point in $(-1, 1)$. Evaluate the claim. (*Evaluate*) [1]

Question 12

Find the equation of the plane passing through the point $(-1, 3, 2)$ and perpendicular to each of the planes $x + 2y + 3z = 5$ and $3x + 3y + z = 0$. (*Apply*) [3]

OR

A line passes through the points $(3, -4, -5)$ and $(2, -3, 1)$.

- (a) Write its equation in Cartesian form. (*Understand*) [1]
 (b) Find the point P where it crosses the plane $2x + y + z = 7$. (*Apply*) [1]
 (c) Find the distance of P from the point $A(3, 4, 4)$. (*Apply*) [1]

Question 13

Using properties of determinants, prove that
$$\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = abc + ab + bc + ca.$$
 [3]

(*Analyse*)

OR

Given $A = \begin{bmatrix} 1 & -1 & 0 \\ 2 & 3 & 4 \\ 0 & 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 2 & -4 \\ -4 & 2 & -4 \\ 2 & -1 & 5 \end{bmatrix}$, find AB . Hence solve the system $x - y = 3$, $2x + 3y + 4z = 17$, $y + 2z = 7$. (*Apply*) [3]

Question 14

Let $I = \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$.

- (i) Show that $I = \frac{\pi}{2} \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} dx$. (*Understand*) [1]
 (ii) Hence evaluate I . (*Apply*) [2]

Question 15

Consider the curve $y = x^2 - 1$ for $0 \leq x \leq 2$.

- (a) Sketch the curve and find where it crosses the x -axis in this interval. (*Understand*) [1]
 (b) Using integration, find the area of the region bounded by the curve, the x -axis and the lines $x = 0$ and $x = 2$. (*Apply*) [2]

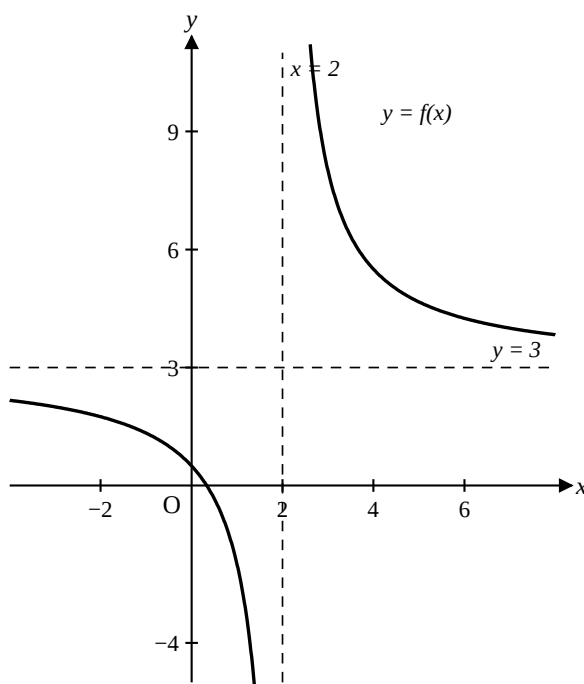
OR

Using integration, find the area of the triangle bounded by the lines $y = 2x + 1$, $y = 3x + 1$ and $x = 4$. (*Apply*) [3]

SECTION D – 25 MARKS

Question 16

A tariff formula is modelled by $f: \mathbb{R} - \{2\} \rightarrow \mathbb{R}$, $f(x) = \frac{3x-1}{x-2}$, whose graph is shown.



- (i) Show that f is one-one. (*Understand*) [1]
- (ii) Find the range of f . Is $f: \mathbb{R} - \{2\} \rightarrow \mathbb{R}$ onto? Give a reason. (*Analyse*) [1]
- (iii) $g: \mathbb{R} - \{2\} \rightarrow \mathbb{R} - \{3\}$ is defined by $g(x) = f(x)$. Find $g^{-1}(x)$. (*Apply*) [1]
- (iv) Verify that $(g \circ g^{-1})(4) = 4$. (*Apply*) [1]
- (v) Find the values of x for which $g(x) = g^{-1}(x)$. (*Analyse*) [1]

Question 17

The height (in hundreds of metres) of a hiking trail above a base camp is modelled by $h(x) = x^3 - 6x^2 + 9x + 1$, $0 \leq x \leq 5$, where x km is the horizontal distance from the camp.

- (a) Find the intervals on which the trail climbs and on which it descends. (*Analyse*) [1]
- (b) Find the points of the trail where it is momentarily level. (*Apply*) [1]
- (c) Find the highest and lowest points of the trail on $[0, 5]$. (*Evaluate*) [2]
- (d) Find the equation of the tangent to the graph of h at $x = 2$. (*Apply*) [1]

OR

An open box with a square base is to be made from a square sheet of tin of side 18 cm by cutting off equal squares of side x cm from each corner and folding up the flaps.

- (a) Express the volume V of the box as a function of x . (*Understand*) [1]
- (b) Find x for which the volume is maximum. (*Apply*) [2]
- (c) Verify that it is a maximum and find the maximum volume. (*Evaluate*) [2]

Question 18

Answer the following.

- (a) Evaluate $\int \frac{dx}{x^2 + 4x + 8}$. (*Understand*) [2]
- (b) Hence, evaluate $\int \frac{2x + 5}{x^2 + 4x + 8} dx$. (*Evaluate*) [3]

OR

- (a) Evaluate $\int x \log x \, dx$. (*Understand*) [2]
- (b) Hence, evaluate $\int \sin 2\theta \log(\sin \theta) \, d\theta$. (*Evaluate*) [3]

Question 19

Bag I contains 3 red and 4 black balls; Bag II contains 5 red and 6 black balls. One bag is chosen at random and a ball is drawn from it.

- (a) Find the probability that the ball drawn is red. (*Apply*) [2]
- (b) The ball drawn is red. Find the probability that it came from Bag II. (*Apply*) [2]
- (c) Hence write the probability that it came from Bag I. (*Understand*) [1]

OR

Two cards are drawn at random, without replacement, from five cards numbered 1, 2, 3, 4, 5. Let X denote the larger of the two numbers drawn.

- (a) Write the possible values of X. (*Understand*) [1]
- (b) Find the probability distribution of X. (*Apply*) [2]
- (c) Find the mean of X. (*Apply*) [2]

Question 20

A fruit seller's receipts on three days read: 2 kg apples, 1 kg grapes and 3 kg oranges for ₹500; 1 kg apples, 2 kg grapes and 1 kg oranges for ₹340; 3 kg apples, 1 kg grapes and 2 kg oranges for ₹560. Prices did not change. Express this as a matrix equation and solve it by the matrix method to find the price per kg of each fruit. (*Evaluate*) [5]

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